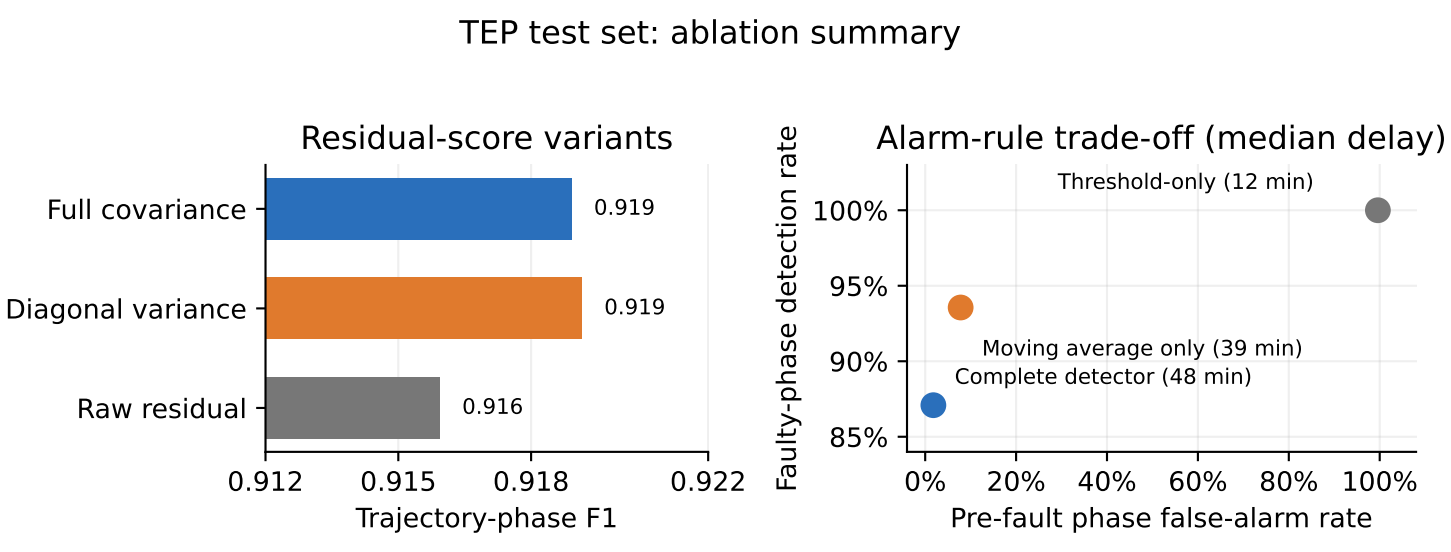
TEP test set: ablation summary
Residual-score variants
Full covariance
Diagonal variance
Raw residual
0.919
0.919
0.916
0.912
0.915
0.918
0.922
Trajectory-phase F1
Alarm-rule trade-off (median delay)
Faulty-phase detection rate
100%
95%
90%
85%
Threshold-only (12 min)
Moving average only (39 min)
Complete detector (48 min)
0%
20%
40%
60%
80%
100%
Pre-fault phase false-alarm rate